Two-mode Network Analysis of Culture Concepts

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Abstract

Kroeber and Kluckhohn's classic 1952 monograph *Culture: A Critical Review of Concepts and Definitions* (Kroeber and Kluckhohn, 1952) brought together over a hundred scholarly definitions of culture and indexed the key concepts they discuss. We build one network from that index to explore how concepts and definitions are connected. In the *definitions network*, concepts like **society** and **beliefs** are linked to 171 definition codes, grouped into seven categories and credited to the author or authors that K&K index under each. Each code is one quotation, identified by its author, year, page numbers and opening words. We present basic statistics about the network, visualize the data using correspondence analysis and spanning trees, and highlight the key concepts and definitions that anchor it. Data-driven types found in the concept profiles and in the wording of the definitions do not recover K&K's categories at better than chance.

1 Data and networks

We build one network from the Index B of Kroeber and Kluckhohn (1952), hereafter K&K. The *definitions network* links 470 concepts—concrete ideas like **society**, **customs**, and **beliefs**—to 171 definitions of culture, which K&K number as codes. Each code is credited to the author or authors K&K index under it, and falls into one of seven categories, which K&K label with the capital letters A through G: A Descriptive, B Historical, C Normative, D Psychological, E Structural, F Genetic, and G Incomplete.[1] Of the 171 codes, 169 appear in K&K's own author index. The other two, C-I-18a and D-II-16, are attributions cited in the main text of the book that the index does not list. Boroch (2016) quotes two parts of four K&K codes separately, lettered (a) and (b): A-3, E-5, F-II-4 and F-II-7. We keep each of these as two codes, A-3(a) and A-3(b) and so on, which is why the eight variant codes replace four. K&K print one index entry per code, so each variant starts with its parent's concept links. For 29 of the 41 concept links copied in this way, we kept the link only on the variant whose quotation contains the concept's name; the other 12 remain on both variants. That assignment is our reading of the quotation text, not taken from anything directly printed in the K&K index. Concepts have consistent names; where a concept appears under different headings in K&K's source text, we treat it as a single concept. Each code is one quotation, identified by its author, year, page numbers and opening words. Section A of the appendix describes how each code was matched to its quotation and how the a and b variants were built.

The 171 × 470 definitions network naturally breaks into 10 separate components, with the largest containing 626 nodes. Our analysis focuses on this largest component, which contains virtually all the meaningful connections. We use maximum spanning trees across multiple projections to identify the strongest connections in different views of the network.

[1] K&K further divide three of the seven categories into numbered subcategories: Normative (C) into C-I and C-II, Psychological (D) into D-I through D-IV, and Genetic (F) into F-I through F-IV. The other four categories have no subcategories. Codes that sit in a subcategory carry its numeral, as in C-I-18a or F-II-4 below.

Table 1: Summary statistics for the definitions network. Statistics describe the full network, before the restriction to the largest component's 2-core used in the analyses. Average nearest-neighbor degree (k_{nn}) is the average degree of a node's neighbors; degree correlation shows how a node's degree relates to the degree of its neighbors. Degree variances are sample variances (divisor $n-1$) over all nodes of each type, including isolates.

	Network
Edges	1201
Density	0.015
Degree correlation r	−0.076
Concepts ($N = 470$)	
Mean degree	2.56
Minimum degree	1
Maximum degree	36
Degree variance	16.34
Mean k_{nn}	10.18
Definitions ($N = 171$ codes, 5 isolated)	
Mean degree	7.02
Minimum degree	0
Maximum degree	25
Degree variance	20.52
Mean k_{nn}	9.22

2 Overview of the network

Table 1 gives summary statistics for the definitions network. The first three rows show node and edge counts. The "degree" rows count connections of each node type; a concept's degree is how many definitions are linked to it in K&K's index, while a definition's degree is how many concepts it is linked to in the index. Concepts and definitions differ substantially in their average degree.

The degree correlation is −0.076, implying that concepts with many connections tend to link to definitions with few, and vice versa, though the association is weak.[2] The nearest-neighbor rows show the other side of this pattern. A concept's neighboring definitions contain 10.2 concepts on average, and a definition's neighboring concepts appear in 9.2 definitions on average, so concepts attach preferentially to heavily used definitions and definitions attach to widely shared concepts (Barabási and Albert, 1999)

Figure 1 shows the full-network degree distributions for the concept nodes (left) and the definition nodes (right). Most concepts are poorly connected: 79% appear in two or fewer definitions, while a small minority of concepts, such as **social**, **society**, **group**, **behavior** or **ideas**, are "super-connectors" appearing in many definitions (see Table 2). This leads to the classic "long tailed" degree distribution for definitions observed in the left-hand side of Figure 1.[3]

[2] Against a degree-preserving null that swaps memberships between random pairs of concepts until the mixing reaches a limit (1000 draws; seed 6128), the null mean is −0.033 with SD 0.022 ($z = -1.94$), and the observed value sits at the 4.1% point of the two-sided null distribution ($p = 0.041$). The negative association is therefore marginally stronger than degree structure alone would produce.

[3] Over the full range of concept degrees, a discrete power law fits better than a truncated exponential (Vuong $z = 6.1$, $p < 0.001$; $\hat{\alpha} = 2.13$), suggesting a mechanism of "preferential attachment" of definitions to already popular

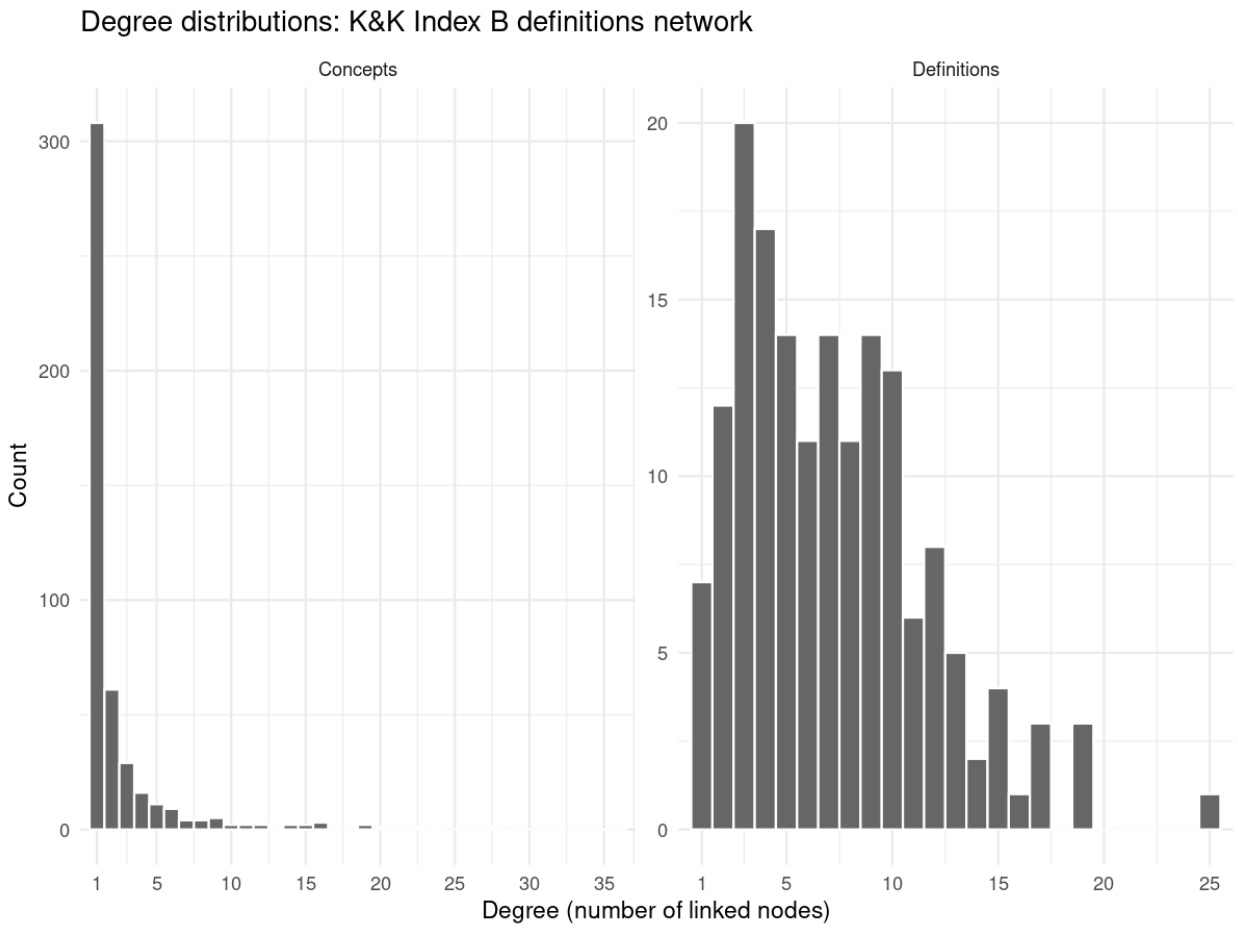


Figure 1: Degree distributions of the definitions network. Left: sub-element (concept) nodes. Right: definition (code) nodes.

Definitions are even more evenly connected, with the average definition covering 5.9 concepts (median 5) and a few covering many more, up to 15 (Table 2).[4] The degree distribution differs from that of the definitions accordingly, coming closer to a bell-shape with a well-behaved (representative) mean. Thus, few definitions are "lean" (relying on a few concepts) and even fewer are conceptually profligate, with most definitions relying on a moderate number of concepts. This means that while most definitions help themselves to a small set of ubiquitous concepts, they also draw from a wider set of concepts not necessarily shared across definitions.

3 Maximum Spanning Trees

By construction, the two-mode definitions network has no ties between two concepts or between two definitions, so we analyze the implied concept-to-concept ties and the definitions-to-definitions

concepts (Barabási and Albert, 1999). The advantage shrinks for degrees of 2 and up ($z = 1.9$, $p = 0.06$) and vanishes for degrees of 3 and up ($|z| < 1$), so the tail alone does not distinguish the two.

[4]Across the 171 definitions, the number of words in each correlates with the number of 2-core concepts the code links to: Pearson $r = 0.55$, Spearman $\rho = 0.57$. Excluding the 22 codes outside the 2-core gives $r = 0.60$ and $\rho = 0.61$ on 149 codes. Word counts use the English translation for the four non-English quotations and exclude Boroch's ellipsis markers.

Table 2: Highest-degree nodes of the definitions network on the LCC 2-core. Concept degree counts the 2-core codes that contain the concept; code degree counts the 2-core concepts the code contains. The last two columns rank codes by degree per 10 words of quotation, using the English translation where one exists. Per-word rankings favor short quotations: across the 171 codes the per-word value correlates −0.23 with word count. Ties at the cutoff are included.

Concepts, by degree		Definitions, by degree		Definitions, by degree per 10 words	
Concept	Degree	Definition (code)	Degree	Definition (code)	Per 10 words
SOCIAL	35	F-I-8 (Reuter, 1939)	15	A-3(a) (Dixon, 1928)	5.00
SOCIETY	31	F-II-8 (Taylor, 1948)	15	D-II-4 (Kluckhohn, 1942)	4.29
GROUP	27	A-8 (Hiller, 1933)	14	D-III-2 (Young, 1934)	4.29
BEHAVIOR	24	C-I-6 (Gillin and Gillin, 1942)	14	F-III-1 (Bain, 1942)	4.29
TOTAL	22	C-II-4 (Bidney, 1946)	13	A-1 (Tylor, 1871)	4.14
IDEAS	21	D-I-7 (Panunzio, 1939)	13	B-13 (Groves and Moore, 1940)	3.75
BELIEFS	19	E-3 (Ogburn and Nimkoff, 1940)	13	G-4 (Rouse, 1939)	3.75
INDIVIDUALS	19	F-I-19 (Murdock, 1949)	13	F-II-6 (Kluckhohn and Kelly, 1945)	3.64
SUM	19	F-I-3 (Folsom, 1931)	13	B-6 (Malinowski, 1931)	3.46
CUSTOMS	16	A-1 (Tylor, 1871)	12	A-3(b) (Dixon, 1928)	3.33
LEARNING	16	C-I-17 (Frank, 1948)	12	F-I-10 (Dodd 1941)	3.33
				F-I-15 (Carr, 1945)	3.33

Note: Degrees are computed on the LCC 2-core used in the analyses (149 codes and 160 concepts). Table 1 and Figure 1 describe the full network instead. The 22 codes outside the 2-core have degree 0 and a per-10-word value of 0. Per-10-word values are code degree × 10 divided by the word count of the quotation. Ties at the cutoff are included, so the three lists have different lengths. F-I-10 is Dodd 1941 as Boroch attributes it, a 1941 committee mimeograph of four authors.

connections via dual one-mode projections (Everett and Borgatti, 2013). Following Breiger (1974), two concepts are tied if they share at least one definition, with weight equal to the number they share. Likewise, two definitions are tied through the concepts they share. In matrix form, the projections are:

$$W_{\text{concept}} = AA^{\top}, \qquad W_{\text{def}} = A^{\top}A, \tag{1}$$

Where $(W_{\text{concept}})_{ij} = \sum_k a_{ik}a_{jk}$ counts the definitions that include both concepts i and j. Each projection is a weighted graph whose edges are the off-diagonal pairs with positive weight. We build projections from the 2-core of the network's largest component.

Projected graphs are typically dense and fully connected making it hard to detect their underlyinng structure. To do so, we extract the "backbone" (Neal, 2014) of each projection by building the maximum spanning tree (MST). The MST of each projection is found by computing the minimum spanning tree on negated weights, $-w$. Kruskal's (1956) method sorts edges from largest weight to smallest and adds each unless it would close a cycle, producing a tree that maximizes $\sum_{e\in T} w_e$. Because the 2-core is connected, each tree has exactly $|V| - 1$ edges.

Many projection weights tie, so a maximum spanning tree is not unique. We therefore build each tree by Kruskal's method with tied edges in a random order, repeat this 1000 times for each projection, and report each node's mean Freeman-normalized (Freeman, 1978) betweenness over those trees, together with its spread and the share of trees in which it ranks in the top ten. The same procedure applies to the concept and definition projections. The tree drawn in each figure is the ensemble member whose betweenness vector is closest to the mean (smallest squared distance). Node size in the figures shows the ensemble mean, so sizes need not match the drawn tree's own betweenness, and node color shows the community structure detected on the drawn tree using Newman's (2006) leading-eigenvector method.

Table 3: Top 10 nodes by mean Freeman-normalized betweenness over 1000 random maximum spanning trees of each projection. Degree is the node's degree in the 2-core. Top-10 share is the percentage of trees in which the node ranks in the top ten. Concepts and definitions are ranked separately. Ties are broken at random in each tree, so the ranks below the first place are not robust: A-10 (10th, share 45%) and A-1 (11th, share 50%) differ by 1.1 points in mean betweenness.

Node type	Node	Author-year	Mean norm. betw. (%)	Top-10 share (%)	Degree
Concepts	GROUP		66.0	100	27
	SOCIAL		57.8	100	35
	SOCIETY		54.7	100	31
	BEHAVIOR		51.6	100	24
	CUSTOMS		36.9	100	16
	BELIEFS		25.7	97	19
	MEMBERS OF A SOCIETY		21.5	99	14
	HABITS		19.3	92	15
	IDEAS		15.8	60	21
	LEARNING		12.5	51	16
Definitions	C-I-6	Gillin and Gillin 1942	77.2	100	14
	A-16	Bidney 1947	35.3	77	11
	F-I-3	Folsom 1931	33.3	86	13
	C-II-4	Bidney 1946	29.1	77	13
	F-I-19	Murdock 1949	28.5	71	13
	C-II-3	Bidney 1942	26.9	71	10
	F-II-8	Taylor 1948	25.9	52	15
	A-8	Hiller 1933	24.1	77	14
	C-I-4	Klineberg 1935	21.8	70	8
	A-10	Linton 1936	21.3	45	11

Note: Definition codes use the format X-Y-Z, where X is a letter (A–G) for K&K's seven definition categories (Descriptive, Historical, Normative, Psychological, Structural, Genetic, Incomplete), Y is a roman numeral for the subcategory, and Z is the number of the definition within it. Variants such as A-3(a) and A-3(b) are the two quotations of one K&K code, and they are counted as separate codes (see the data section). The author–year column names the source of each definition's quotation, as Boroch attributes it; the same author and year label the node in Figure 3.

The concept-level tree (Figure 2) shows how 160 concepts cluster in the concept projection and the definition-level tree (Figure 3) shows how 149 definitions cluster in the definitions projection. Table 3 lists the top 10 concepts and definitions by mean betweenness over the tree ensembles, with each node's top-ten share and 2-core degree.

Table 3 reports two kinds of measure for each node. Mean normalized betweenness is the share of the paths in a tree that run between other pairs of nodes and pass through the node, averaged over the 1000 trees and expressed as a percentage of the largest value a node could take in a tree, which is the center of a star. It indexes how much a node sits between otherwise separate parts of the projection. The top-10 share is the percentage of trees in which the node ranks in the top ten, so it measures how much the node's place depends on the arbitrary tie-breaking; a high mean with a low share is a tree-dependent result. Degree is a different quantity, the count of a node's ties in the 2-core, and the two can diverge. **customs** has 16 ties and more than twice the betweenness of **ideas**, which has 21, while **learning** has 16 ties and only 12.5%. The same holds among definitions: C-I-6 has 14 ties but lies on 77% of the maximum, the largest share in the ensemble, which suggests it links regions of the tree that its degree alone would not reveal. The six highest-ranked concepts stay in the top ten in at least 97% of trees, so their places are robust, whereas A-10, in tenth place

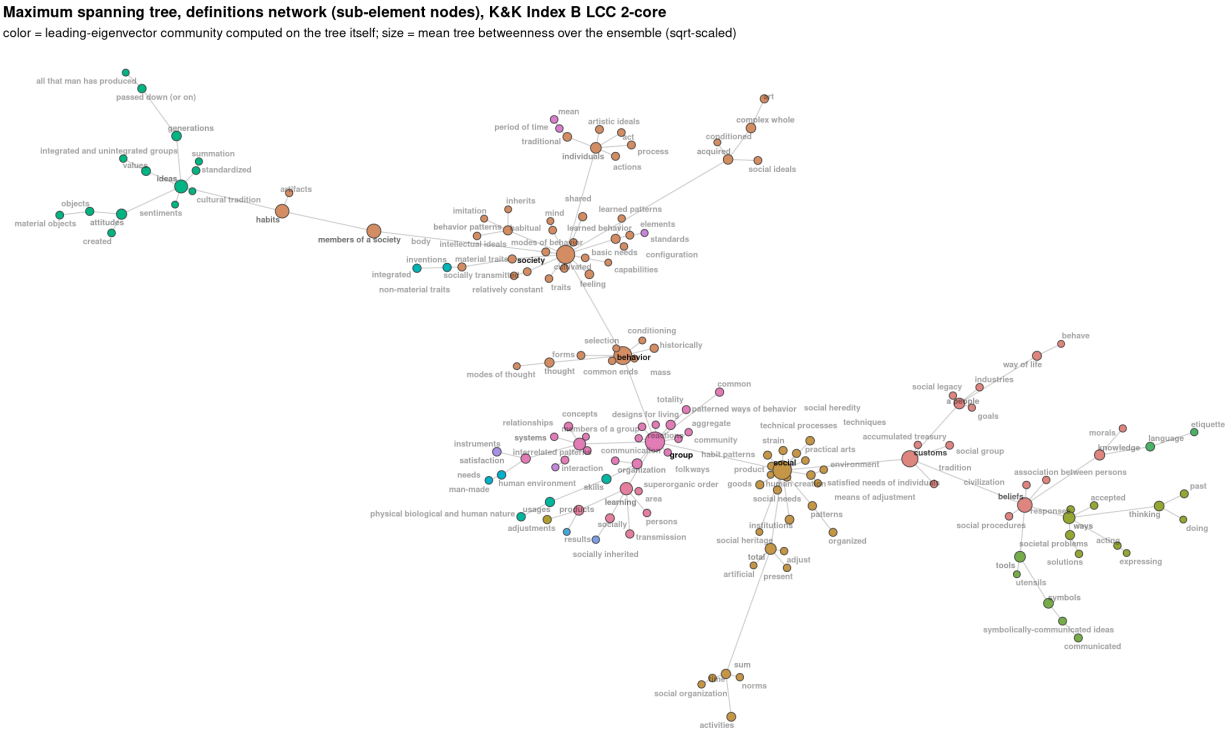


Figure 2: Maximum spanning tree of the definitions network, sub-element (concept) nodes. Color is the leading-eigenvector community on the drawn tree; size is mean betweenness over the tree ensemble.

with a 45% share, is at the cutoff and could be replaced by A-1 in other draws. Substantively, the concepts at the top are the vocabulary that most often joins the definitions' regions in the tree, while the definitions with lower shares occupy positions that depend on which equally weighted tree is drawn.

3.1 Concept Tree

In the concept tree (Figure 2), the six concepts with the highest mean betweenness—**group**, **social**, **society**, **behavior**, **customs**, and **beliefs**—are the top six of Table 3. **group** is the hub of a pink region of **communication**, **organization**, **folkways**, **systems**, **designs for living**, and **reactions**. **social** heads a gold region of social-organizational vocabulary (**social heritage**, **institutions**, **social needs**, **patterns**, **goods**, **human creation**), and **customs** heads a salmon region with **a people**, **accumulated treasury**, **social group**, and **tradition**. **society** heads a large orange region with **individuals**, **shared**, **socially transmitted**, **learned behavior**, and **basic needs**, and **behavior** heads the orange region below it, with **thought**, **forms**, **historically**, **selection**, and **common ends**. **beliefs** heads a green region of **ways**, **tools**, **knowledge**, **civilization**, and **association between persons**. **ideas** heads a teal branch to the upper left, with **attitudes**, **sentiments**, **values**, **generations**, and **cultural tradition**; **habits** links that branch to **members of a society** and so to **society**. The acquisition and transmission vocabulary of K&K's "Genetic" definitions sits on a peripheral branch: **learning** hangs off **group**, with **transmission**, **socially**, **products**, and **persons** attached to it, so these concepts reach the core only through **group**.

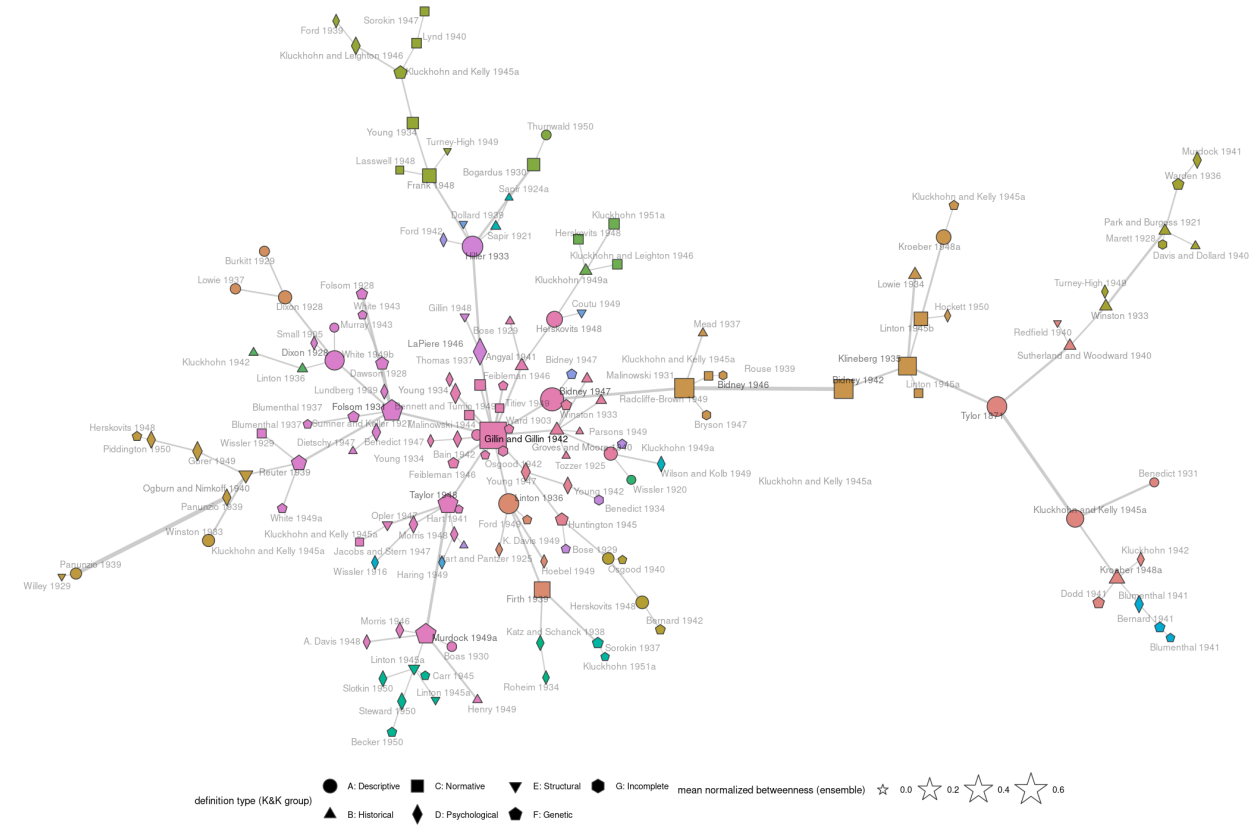


Figure 3: Maximum spanning tree of the definitions network, code (definition) nodes labeled by the author and year of their quotation. Shape is the K&K definition group (A–G, Descriptive to Incomplete); color is the leading-eigenvector community on the drawn tree; size is mean betweenness over the tree ensemble.

The concept tree is mostly a backbone of low-betweenness nodes. Averaged over the 1000 trees, 21 of the 160 concepts are leaves in every tree, and the median concept is a leaf in 76% of them. Degree is the number of 2-core codes containing a concept. It is positively associated with mean betweenness (Spearman $\rho = 0.71$), but not uniformly. The four highest-degree concepts, **social** (35), **society** (31), **group** (27), and **behavior** (24), are among the six highest in mean betweenness. Other high-degree concepts bridge little: **total** (degree 22) has mean betweenness 7.3%, **ideas** (21) 15.8%, **sum** (19) 2.6%, and **transmission** (15) 0.9%. Element names repeat across headings, and **sum** and **total** both appear under the totality heading, so these counts may combine distinct elements.

Two concepts at the edges of this tree need separate mention. **a people**—an alternative reference to a group or collective—links the salmon **customs** region to **way of life**, and removing it separates **way of life** (with **behave**) from **customs**. Its strongest ties are to **way of life** (six shared definitions) and **customs** (five), and its fifteen definitions come from all seven groups, so it is a broadly shared concept rather than one tied to a single group. This speaks to the penchant to define culture as the customs or way of life common to "a people." **complex whole**, the classic locution due to the "original" definition by Tylor (1871), heads a small cluster at the upper right of the figure. Its tree neighbors are **acquired** and **art**. The cluster reaches **society** only

through **acquired**, and its six definitions come from Groups A (Descriptive), B (Historical) and D (Psychological).

3.2 Definitions Tree

The three codes with the highest mean betweenness are the Gillin and Gillin (1942) quotation (C-I-6, 77%), the Bidney (1947) quotation (A-16, 35%), and the Folsom (1931) quotation (F-I-3, 33%). The Gillin quotation is the largest node in the figure. Gillin and Gillin (1942) heads the largest community (18 definitions, drawn from six of the seven groups), and Bidney (1947) sits in the same community. The Gillin quotation is wordy (64 words) but not among the wordiest (89th percentile of length among codes with a quotation), so its bridging role is not a simple function of its length but of the concepts it uses. Here is the definition in full, with the concepts it uses in bold:

> The **customs**, **traditions**, **attitudes**, **ideas**, and **symbols** which **govern social behavior** show a wide variety. Each **group**, each **society** has a set of **behavior patterns** (overt and covert) which are more or less **common** to the members, which are passed down from **generation** to **generation**, and **taught** to the **children**, and which are constantly liable to **change**. These **common** patterns we call the culture...

The two trees' betweenness rankings are each their own structural fact rather than echoes of one another. A concept's mean betweenness is only weakly related to the mean betweenness of the codes that contain it (Pearson $r = 0.13$; Spearman $\rho = 0.30$), and the reverse relation is weaker still ($r = 0.08$; $\rho = 0.10$). Concepts gain concept-tree betweenness by bridging concept regions, not by riding on definitions with high definition-tree betweenness: **group** owes its top concept-tree rank to its position among concepts, not to its definition partners. The converse also appears. A-10 (Linton 1936) ranks tenth in definition betweenness with only **society** among the six highest in concept betweenness, so a definition can bridge definitions without containing any bridge concept. The apparent association between the top concepts and the top code is largely one hub: five of the six concepts with the highest concept betweenness (**group**, **social**, **society**, **behavior**, and **customs**) link to C-I-6, which has the highest mean betweenness among the codes of each of them. The sixth, **beliefs**, does not link to it; its highest-betweenness code is F-I-3.

C-I-6 has the highest mean betweenness in the definition tree (77%), and its position is not a simple function of its length. At 64 words it sits at the 89th percentile of quotation length among codes with a quotation, well short of the longest quotation (142 words). It links to 14 concepts, and the mean concept betweenness of those concepts sits at the 89th percentile of codes. Its degree is not the highest: F-I-8 and F-II-8 each link to 15 concepts. But a high-betweenness vocabulary is not sufficient: 93 other codes share at least one of its five highest-betweenness concepts, and their mean betweenness is 4.7% against C-I-6's 77.2%. What is distinctive about C-I-6 is where it sits in the definition tree: it bridges definitions, and it does so with a concept repertoire that many other definitions share.

Bidney's (1947) is in many ways more impressive. It ranks second in mean definition betweenness (35%), but it is short (only 34 words). Its links are a mix. Of its 11 concepts in the 2-core, three are among the six highest in concept betweenness (**social**, **behavior**, and **society**), and the mean concept betweenness of its linked concepts sits at the 68th percentile. It has one further link outside the 2-core. Its position in the drawn tree is what gives it high betweenness: removing it leaves components of 116 codes, 31 codes, and one code. Here is the definition in full. Of its linked concepts, only *acquired* and *society* appear verbatim in the text; **behavior** is linked but does not appear in the quotation:

> Culture is that complex whole which includes artifacts, beliefs, art, all the other habits **acquired** by man as a member of **society** and all products of human activity as determined by these habits.

Two other Bidney definitions, C-II-4 and C-II-3, share a community that contains neither C-I-6 nor A-16. First, we have C-II-4 (Bidney, 1946), the fourth highest mean definition betweenness (29%). It is a "Normative" definition according to K&K, and it shares its community with C-II-3 (Bidney, 1942) but not with C-I-6 (Gillin and Gillin, 1942) or A-16 (Bidney, 1947). Its quotation is 40 words long, and it links to 13 concepts, among them **behavior** (fourth highest concept betweenness) and **society** (third):

> An integral or holistic concept of culture comprises the **acquired** or **cultivated behavior**, **feeling**, and **thought** of **individuals** within a **society** as well as the **patterns** or **forms** of intellectual, social, and artistic ideals which human societies have professed **historically**.

Finally, we have C-II-3 (Bidney, 1942), the sixth highest mean definition betweenness (27%). It is also a "Normative" definition according to K&K. Its quotation is 40 words long, and it links to 10 concepts in the 2-core, among them **society** (third), **behavior** (fourth), and **members of a society** (seventh):

> A culture consists of the **acquired** or **cultivated behavior** and **thought** of **individuals** within a **society**, as well as of the intellectual, artistic, and social ideals which the members of the society profess and to which they strive to conform.

Folsom (1931) (F-I-3), the third highest in mean definition betweenness (33%), is a Genetic definition that reaches the top ten in 86% of trees. Its 75 words put it at about the 95th percentile of length among codes with a quotation, and it links to 13 concepts in the 2-core (16 in the full network). Its strongest link is to **beliefs**, which is among the six concepts with the highest mean concept betweenness. Its other concepts range from **tools**, **attitudes** and **symbols** to **generations**, **common** and **sum**, several of which have low concept betweenness. Unlike the Gillin community, which draws on six of the seven groups, the drawn definition tree places it in a community of 15 codes, eight of them Genetic, so it sits mostly with its own group. Its high betweenness therefore does not come from bridging groups in the same way; the tree-level evidence points to its link to **beliefs** and its breadth of concepts, but we have not decomposed its betweenness path by path, and its tree neighbors should be checked on the drawn figure.

Murdock (1949) (F-I-19) has the fifth highest mean definition betweenness (29%). Its 70 words link to 13 concepts in the 2-core, three of which are among the four concepts with the highest concept betweenness (**group**, **behavior**, and **society**), and it has six tree neighbors, among them Taylor (1948) (F-II-8) and E-5(b). Its quotation frames culture as the product of a process, the "interaction of learning and society":

> The **interaction** of **learning** and **society** thus produces in every human **group** a **body** of **socially transmitted adaptive behavior** which appears super–individual because it is **shared**, because it is **perpetuated** beyond the individual life span, and because its quantity and quality so vastly exceeds the capacity of any single person to achieve by his own unaided effort. The term culture is applied to such **systems** of **acquired** and transmitted **behavior**.

Kroeber's A-17 (65 words; pp. 8–9) ranks 25th by mean betweenness (5.7%) and links to 10 concepts in the 2-core. Only one of them, **behavior**, is among the four highest in concept betweenness. The others include **habits** (19.3%), **ideas** (15.8%), and **learning** (12.5%), while **mass** has no betweenness in any tree and **techniques**, **totality**, and **transmission** are below 1.2%. Its tree neighbors are C-I-8 and F-II-6. Its quotation lists the components of culture and then frames it as a total product:

> [. . .] the **mass** of learned and transmitted **motor reactions**, **habits**, **techniques**, **ideas**, and **values** – and the **behavior** they induce – is what constitutes culture. Culture is the special and exclusive product of men, and is their distinctive quality in the cosmos [. . .]. Culture [. . .] is at one and the same time the **totality** of products of social men, and a tremendous **force** affecting all human beings, **socially** and **individually**.

Seven of the top ten codes carry a C (Normative) or F (Genetic) code; the other three, A-16, A-8, and A-10, are Descriptive.

Authors supply several of the top-ten codes. Bidney holds three (A-16, C-II-4, and C-II-3), and Linton holds one (A-10). No Kroeber code is in the top ten: A-17 ranks 25th.

The works themselves barely cluster. The 149 codes of the tree carry 120 distinct author–year labels—a repeated label is a second quotation from the same work—and the 19 labels that appear two or more times almost never stay together: 15 of them spread over two or more communities. Kluckhohn and Kelly (1945)'s chapter, the work with the most codes in the tree (seven), reaches five communities, and Herskovits (1948) spans four. Of the four codes split into variants, E-5, F-II-4 and F-II-7 each keep both variants in one community (the three communities differ). The two A-3 variants (Dixon, 1928) fall in different communities. Linton (1945)'s third code (C-I-9) is in another community from its two E-5 variants. The regions of the tree organize concept vocabulary, not the works the definitions were drawn from.

The six concepts with the highest mean concept betweenness, **group**, **social**, **society**, **behavior**, **customs** and **beliefs**, are the top of Table 3, and each anchors a region of the concept tree. The definition with the highest mean betweenness is C-I-6, at 77% of the maximum possible, a single Normative code. In the concept tree the acquisition and **transmission** vocabulary sits on a peripheral branch rather than at the core, suggesting that Genetic concerns, however common in the definitions themselves, connect to the rest of the conceptual structure only through a narrow path.

4 Data-driven types: non-negative matrix factorization

The spanning trees and the community colors above classify nodes hard: each code belongs to exactly one community of the tree. Non-negative matrix factorization (NMF) (Lee and Seung, 2001) offers a soft, parts-based alternative. It approximates a non-negative matrix V by a product WH with all entries non-negative, so every code carries a loading on each factor and every factor is an additive mixture of the original features, with no subtractive terms. We ask how far data-driven types agree with K&K's own seven categories A–G (Descriptive to Incomplete), first from the concept profiles of the definitions—which concepts each definition uses—and then from the wording of the definitions. Both factorizations use the same base-R implementation throughout (Euclidean multiplicative updates, deterministic given a seed, with ten random restarts per value of k), and both fix k at seven, one factor per K&K group. Fixing k this way is a choice, not a result: the reconstruction error falls monotonically in k without an elbow, and restart agreement

is irregular (mean adjusted Rand index 0.51 to 0.67 over $k = 3$ to 8 for the concept profiles, 0.48 to 0.70 for the wording), so neither criterion selects a number of factors on its own.

4.1 Types from the concept profiles of the definitions

The first factorization asks the question of the network itself. Its matrix is the definitions network's own biadjacency on the largest component's 2-core: 149 definition codes, each described by the 160 concepts it uses, coded 1 when the definition contains the concept and 0 otherwise. The factorization is run at $k = 7$, one factor per K&K group.

The factors read as recognizable concept profiles rather than K&K's categories (Figure 4), and we label them by their leading concepts: *social organization* (**social**, **social heritage**, **institutions**, **patterns**), *group and way of life* (**group**, **behavior**, **common**, **a people**), *norms and knowledge* (**beliefs**, **customs**, **ways**, **knowledge**), *society and acquired behavior* (**society**, **acquired**, **members of a society**, **individuals**), *transmitted ideas* (**ideas**, **habits**, **generations**, **attitudes**), *complex whole* (**total**, **sum**, **tools**, **activities**), and *systems and skills* (**systems**, **organization**, **satisfaction**, **skills**). The last factor is diffuse: its leading loadings are an order of magnitude smaller than the others', so its label is the least secure. Against the A–G groups (Descriptive to Incomplete) the agreement is indistinguishable from chance (adjusted Rand index 0.002, against a permutation 95th percentile of 0.016), and no factor puts more than 40% of its codes in one group. The subgroup-level view is in the appendix (Figure 6).

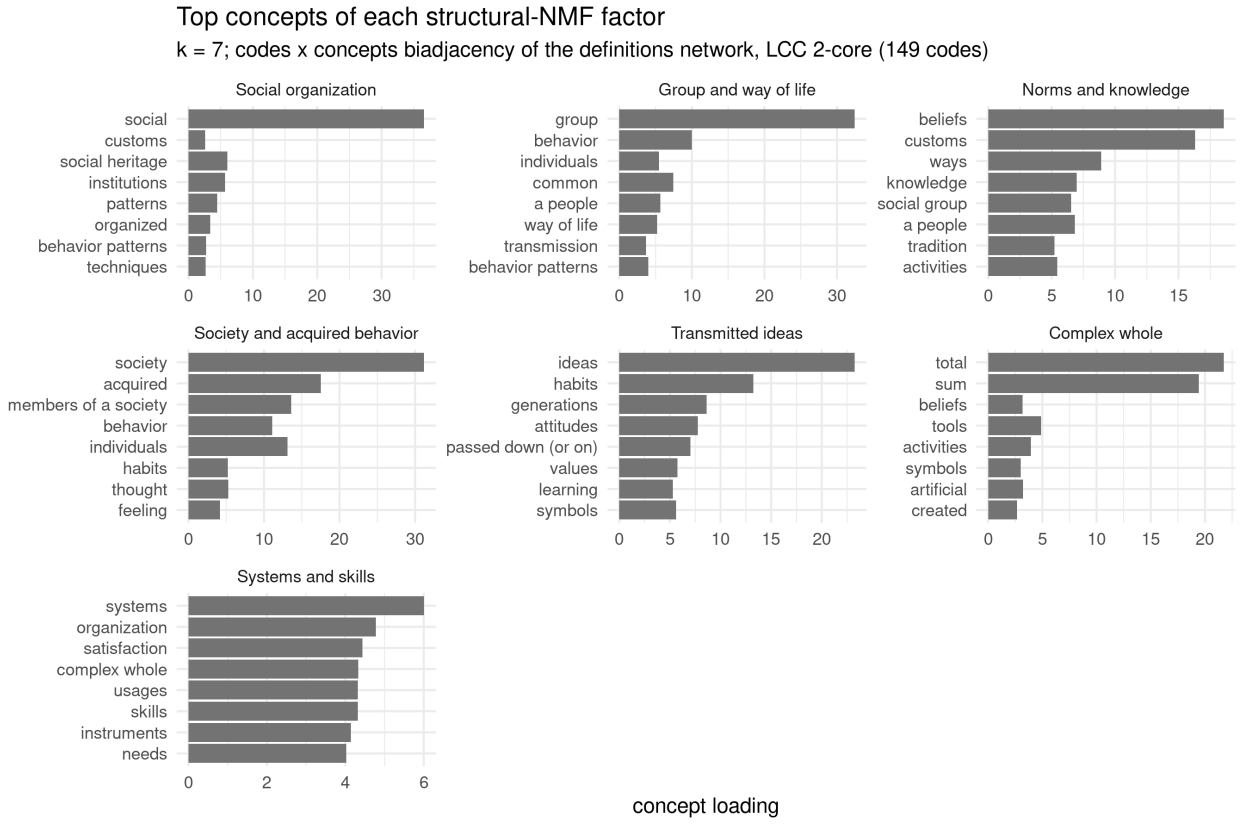


Figure 4: Top concepts of each structural-NMF factor at $k = 7$, from the codes × concepts biadjacency of the definitions network's largest-component 2-core (149 codes).

4.2 Types from the wording of the definitions

The second factorization asks the question of the definition texts themselves, independently of both K&K's coding and the concepts the definitions use.

The second factorization asks the question of the definition texts themselves, independently of both K&K's coding and the concepts the definitions use. The corpus holds 171 definition codes carrying a quotation: the texts from Boroch's Table 2 mapped to K&K codes through the crosswalk, the appendix-only texts, and one statement K&K prints itself (a Part II page footnote). Three Table 2 columns that repeat another column's quotation word for word are not counted again. Documents average about thirty-four words (median 28), so the term–document matrix is built restrictively: tokens of three letters or more, a stopword list that removes the words nearly every definition contains (culture, society, social, man, and the general definitional vocabulary total, whole, learned, acquired), and terms kept to a document frequency between two documents and half the corpus, leaving 403 terms weighted by TF–IDF. The four quotations K&K and Boroch print in French or German enter as English translations (K&K's own rendering for Maquet, ours for the other three, not yet checked against the printed pages). The factorization is run at $k = 7$, one factor per K&K group.

The factors again find recognizable themes rather than K&K's categories (Figure 5), and we label them by their leading terms: *species-specific ability* (events, species, use, dependent, ability), *order and operation* (order, operation, self, superorganic, complex), *mental and holistic* (concept, mean, holistic, traits, mental), *goals and persons* (certain, rather, others, persons, goals), *symbolic process and adjustment* (concrete, process, symbolic, experience, adjust), *material and transmitted behavior* (material, non, transmitted, socially, institutions), and *passed down generations* (generation, passed, down, everything, common). On a corpus this small a factor can be driven by one author's phrasing, so the labels should be read as descriptive of the top terms, not as clean concepts. The labels for mental and holistic and goals and persons are the least clear, since their leading terms include generic words (rather, certain, some, mean). Against the A–G groups (Descriptive to Incomplete) the agreement is close to chance (adjusted Rand index 0.012, against a permutation 95th percentile of 0.016), and no factor has more than 54% of its codes in one group.

4.3 Interpretation

Neither the concept profiles nor the wording of the definitions recovers K&K's taxonomy. The two factorizations fail on different matrices—the binary concept codings of the network and the texts of the quotations—so the null result does not hinge on one representation. Two readings are possible. One is that K&K's groups classify the emphases of individual *statements*, and that these emphases cannot be recovered from a binary concept coding, or from quotations of about thirty words each. The other, which we find more plausible, is that the seven groups are editorial categories—useful for organizing a bibliography—rather than dimensions along which the corpus varies. This is consistent with the spanning-tree picture above, where the vocabulary the Genetic group concerns itself with sits on a peripheral chain of the concept tree rather than forming a region of its own, and with the community structure of the trees, which follows concept vocabulary, not authorial schools. The null result should be read against the method's limits as well: the number of factors was fixed to match the taxonomy under test rather than selected by the data, the short corpus makes TF–IDF factors fragile (restart agreement ranges from 0.48 to 0.70 for k from 3 to 8), and the quotations are the passages Boroch selected, not the authors' full works.

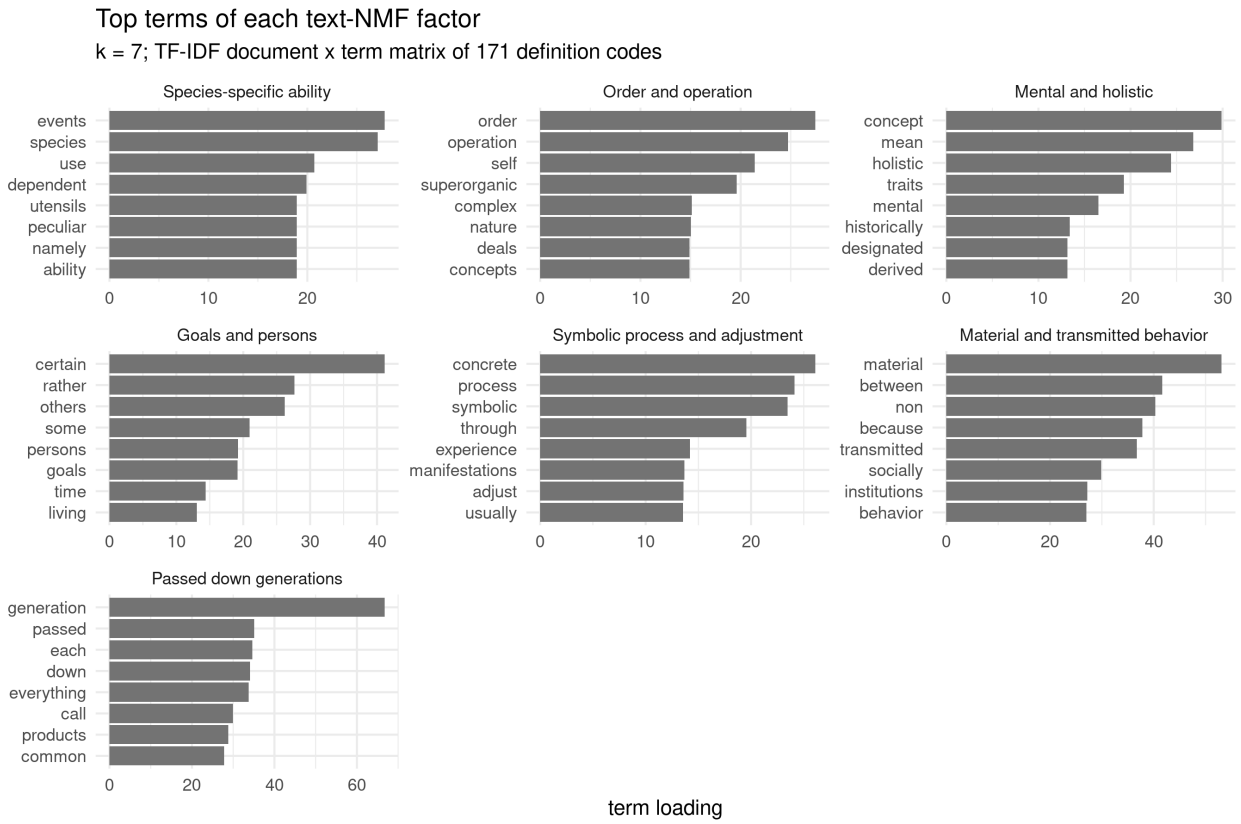


Figure 5: Top terms of each factor of the text NMF at $k = 7$, from the TF–IDF matrix of the 171 definition codes.

5 Limitations

The networks depend on the K&K Index B transcription and on the verified-code filter. Element names are not unique across headings, so same-named elements are merged, which can join unrelated entries. Several readings depend on OCR corrections recorded in the repository's manual-fix files; the validators check structure and round-trips, not the content of each edge. Two codes that are not in Index A are included: C-I-18a and D-II-16, from the main text. K&K's Part III statements (C-5, D-5, D-7, E-17) are excluded. The basis for each is documented in the repository. The four split codes and the 29 link decisions that separate their variants (see the data section) are judgments: the variants share one K&K index entry, and the text-based assignment of concept links is a reading that a reader can check against the quotations, not a printed fact. The definition-level trees and the text factors depend on these decisions. Each projection has many maximum spanning trees, because its edge weights tie heavily. Betweenness averages over 1000 random trees, but the drawn trees show one representative member, so tree-level structure (neighbors, communities and regions) describes that member, and the ranks below the first place in Table 3 vary across trees.

A Constructing the code-level data

The analyses use one object: a table with a row for each of the 171 K&K definition codes. Each row gives the code's group and subgroup, its sub-elements, and the citation that identifies its quotation. This section records how that table was built, the choices made along the way, and the basis for each. Two sources are joined. K&K's Index A and Index B, read from a scan of the 1952 monograph, give each code its author, year and sub-elements. Boroch's Boroch (2016) gives each code a quotation with page numbers. Every join follows a stated rule, and each judgment is recorded with its basis in the project's provenance log. The analyses use one object: a table with a row for each of the 171 K&K definition codes. Each row gives the code's group and subgroup, its sub-elements, and the citation that identifies its quotation. This section records how that table was built, the choices made along the way, and the basis for each. Two sources are joined. K&K's Index A and Index B, read from a scan of the 1952 monograph, give each code its author, year and sub-elements. Boroch's (2016) article gives each code a quotation with page numbers. Every join follows a stated rule, and each judgment is recorded with its basis in the project's provenance log.

A.1 Sources and the unit

The unit of analysis is the code K&K attached to each definition, which corresponds to one printed index entry in K&K, and one quotation in Boroch (2016). Boroch's appendix has 171 codes. Boroch's Table 2 has 135 columns, 132 of which match an appendix code. The two sources number codes differently. Boroch renumbers within each group, so his A10 (Winston, 1933) is K&K's A-9. Codes are therefore matched by author, year, group and subgroup. Positional agreement is kept only as a check: of 100 matches made by name and year, 48 agree with the position.

A.2 Index A and Index B

Index A (authors) and Index B (sub-elements) were read from the OCR text layer and corrected by explicit rules. The corrections are recorded in a manual-fix table of 142 rows, each with its printed-page evidence. A further 10 rows recover codes that the OCR glued into element names, and 14 rows record name decisions. The project avoids general character rules, because a rule such as reading a y as a 5 silently produced codes from unrelated tokens. Each manual fix is keyed to a parsed token, and the checks confirm that every key matches one.

The universe is 171 codes. Of these, 169 appear in K&K's author index. The other two, C-I-18a (Maquet 1949) and D-II-16 (Aberle et al. 1950), are cited in main-text footnotes but not listed in the index; their Index B entries are read from the sub-element entries that carry them. Four Part III statement codes (C-5, D-5, D-7 and E-17) are excluded, since they are K&K's statements rather than Part II definitions.

A.3 Matching Boroch's codes to K&K's

The crosswalk matches each column of Boroch's Table 2 to a K&K code. Of the 135 columns, 117 match automatically when exactly one code fits, and 15 more are settled by hand decisions, each with a basis. Three of Boroch's Table 2 columns have no appendix entry of their own and carry no K&K code: A10a, C1-5a and F4-6. Each is a verbatim copy of another quotation, which leaves 132 columns with a K&K code. A10a repeats A10 (Winston, 1933), C1-5a repeats C1-5 (Firth, 1939), and F4-6 repeats F4-5, which is the entry Boroch's numbering misprints as F4-6. None of the three adds a quotation, so the object does not use them. The F4-5 quotation is Kluckhohn and

Table 4: The four split codes and their variants.

K&K code	Variant	Boroch code and source	Page numbers
A-3	A-3(a)	A3, Dixon 1928	3
A-3	A-3(b)	A4, Dixon 1928	3
E-5	E-5(a)	E5, Linton 1945 (footnote 5)	5
E-5	E-5(b)	E6, Linton 1945	32
F-II-4	F-II-4(a)	F2-4, Blumenthal 1937 (footnote 4)	3
F-II-4	F-II-4(b)	F2-5, Blumenthal 1937	12
F-II-7	F-II-7(a)	F2-8, Feibleman 1946 (tentative; footnote 7)	75
F-II-7	F-II-7(b)	F2-9, Feibleman 1946 (final)	76

Kelly 1945 (printed p. 87), and it gives K&K's F-IV-5. K&K's printed codes A-10a (Lowie, 1937) and C-I-5a (Lynd, 1940) take their quotations from the appendix entries A12 and C1-6.

A.4 Quotations and citation keys

Each code has one quotation. Boroch supplies 170 of them: the 132 codes from Boroch's Table 2 and the 39 appendix-only codes, less the Sears addendum (C1-22), which is linked to a specialty code rather than to a network code. D-III-4 has no Boroch entry, and its quotation is K&K's printed footnote 14 (Roberts 1951, pp. 3 and 6, printed p. 60). Four quotations are not in English and are rendered from a translation table: B-17 and C-I-18a from French, and F-I-5 and F-II-3 from German. C-I-18a uses K&K's own printed English. The other three are working translations that have not been checked on the page images.

Each code is identified by a citation key: Boroch's author, year and page numbers, together with the first six words of the quotation. The key is unique for all 171 codes, with no missing values. The page numbers are Boroch's as printed and have not been checked against the source works. The key therefore identifies a quotation as Boroch reports it, and the page numbers are the first thing to check against a source.

A.5 Splitting the a and b quotations

Boroch quotes parts of four K&K codes separately, labeled (a) and (b). K&K prints one index entry for each of these codes, so the object had one code where Boroch has two quotations. On 2026-10-30 the project split each of the four codes into two variants (Table 4). The split follows Boroch's quotations and is not a distinction K&K prints. Each variant takes its own quotation and its own citation key.

Because K&K gives one index entry per code, each variant starts with a copy of its parent's sub-element links, 41 in all. A lexical screen then assigns 29 of the 41 to one variant. A link stays on the variant whose quotation contains the sub-element's name and not on the other. The 12 links where both or neither quotation contains the name remain on both variants. The screen is a word match, so a name used in a generic sense, such as "total" or "past," can be assigned wrongly. Each assignment is a reading of the quotation text and should be treated as a judgment. Removing a link also removes the heading edge that carries it, which accounts for 34 edge rows. Together the split takes the verified network from 167 to 171 codes and from 1189 to 1201 edges.

A.6 Checks and open items

The build's validators check the network's structure, its round trips and its attributes, but not the content of each edge. The block check places 1301 of the 1326 Index B edge rows (before the 34-row removal) in their printed block, and 25 in another block. The page-image review covers 23 of those 25. Several items remain open. The year of K&K's footnote 5, which Boroch gives as Linton 1945a, was read from the page image at 150 dpi, and a higher-resolution check is still needed. The three duplicate columns need recorded decisions in the crosswalk, and the text analysis must be brought into line with them. The three non-English translations need checking against the page images. The Boroch page numbers need checking against the source works. The 29 link assignments are judgments on the text, as described above.

B Supplementary sankey diagrams at the subgroup level

The NMF comparisons in the body classify the definitions against K&K's seven group letters, first by concept profile and then by wording. The figures here repeat both at the finer grain of K&K's own typology, the roman-numeral subgroups the classification prints within groups C (Normative), D (Psychological), and F (Genetic); groups A (Descriptive), B (Historical), E (Structural), and G (Incomplete) number their codes plainly and have no subgroup level. Each ribbon takes the color of its origin node's group, and node size is the number of codes. The structural panel covers the 149 codes of the largest component's 2-core; the text panel, all 171 codes carrying a quotation.

On the left, A (Descriptive), B (Historical), D-I and F-I reach all seven structural factors, and C-I and D-II reach six. On the right, *group and way of life* and *norms and knowledge* are the largest factors at 26 codes each, and *systems and skills* the smallest at 10. The largest cells are A (Descriptive) on *norms and knowledge* (9 of 22 codes), B (Historical) on *social organization* (7 of 22), and F-I on *complex whole* (6 of 17). No subgroup's codes sit on a single factor; the smallest subgroups in the 2-core, C-II, D-III, D-IV and F-IV (two to four codes each), reach between two and three factors.

On the left, A (Descriptive), C-I and D-II reach all seven factors, and B (Historical), D-I, F-I and F-II each reach six. On the right, *material and transmitted behavior* is the largest factor at 43 codes and draws on D (Psychological, 10), B (Historical, 7) and F (Genetic, 7). *Passed down generations* (34 codes) draws most on D (Psychological, 11) and A (Descriptive, 8), and *symbolic process and adjustment* (26 codes) on F (Genetic, 12 codes, 46% of the factor), B (Historical, 5) and D (Psychological, 4). *Order and operation* (21 codes) draws on F (Genetic, 7), D (Psychological, 4), and A (Descriptive), B (Historical) and C (Normative), 3 each. *Species-specific ability* and *mental and holistic* (17 codes each) both draw on F (Genetic, 6) and C (Normative, 5). *Goals and persons* (13 codes) is mostly D (Psychological, 7 codes, 54% of the factor).

For the definition codes no aggregation is needed, since each code carries its own subgroup. The largest cell holds eight codes (A (Descriptive) on *passed down generations*, 8 of 22 codes), and two cells hold seven (F-I on *symbolic process and adjustment*, 7 of 21; B (Historical) on *material and transmitted behavior*, 7 of 23). No subgroup's codes sit on a single factor. The smallest subgroups, C-II (6 codes), D-III (4) and D-IV (2), reach between two and four factors.

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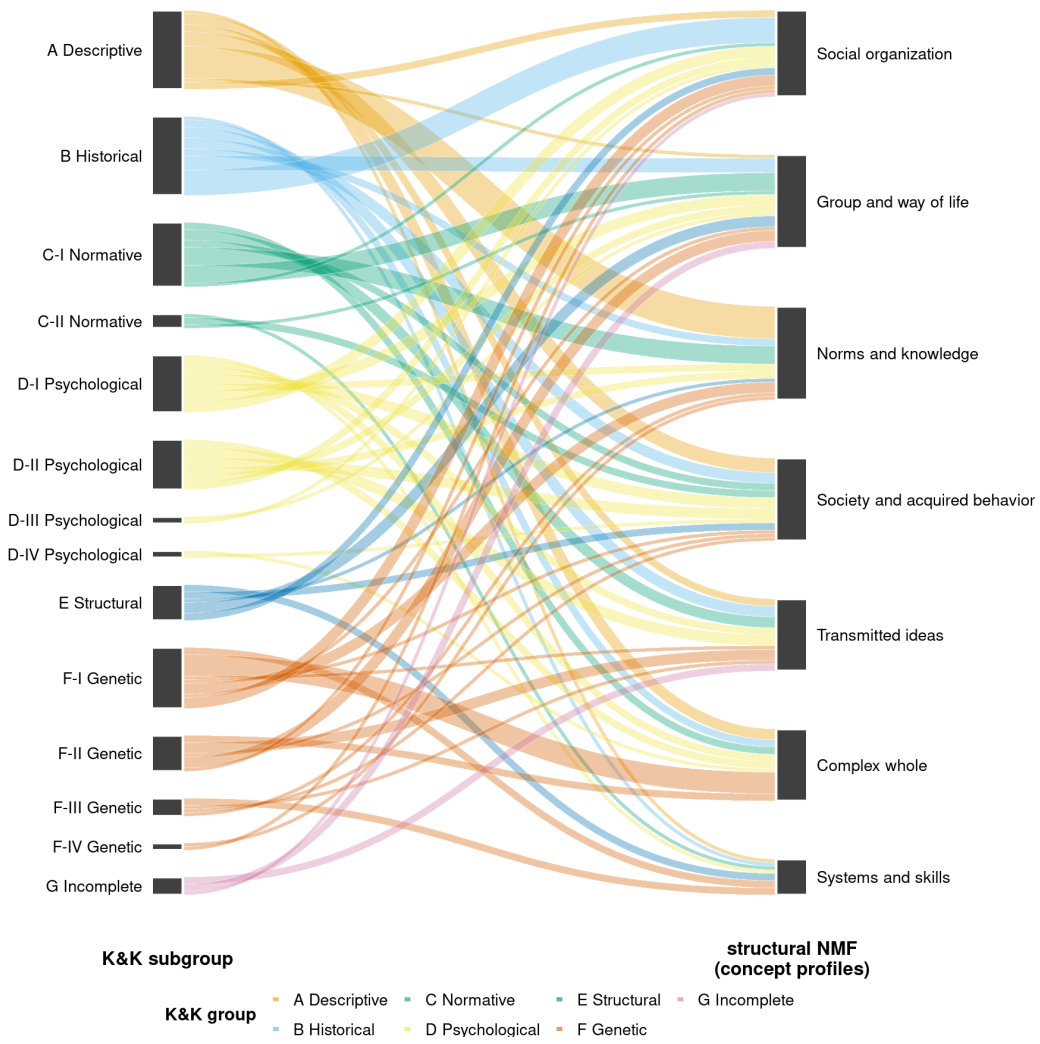


Figure 6: The 149 definition codes of the largest component's 2-core, by their own K&K subgroup (left) and dominant structural-NMF factor (right). Ribbons take the color of their origin node's group.

Barabási, A.-L. and Albert, R. (1999). Emergence of scaling in random networks. *Science*, 286(5439):509–512.

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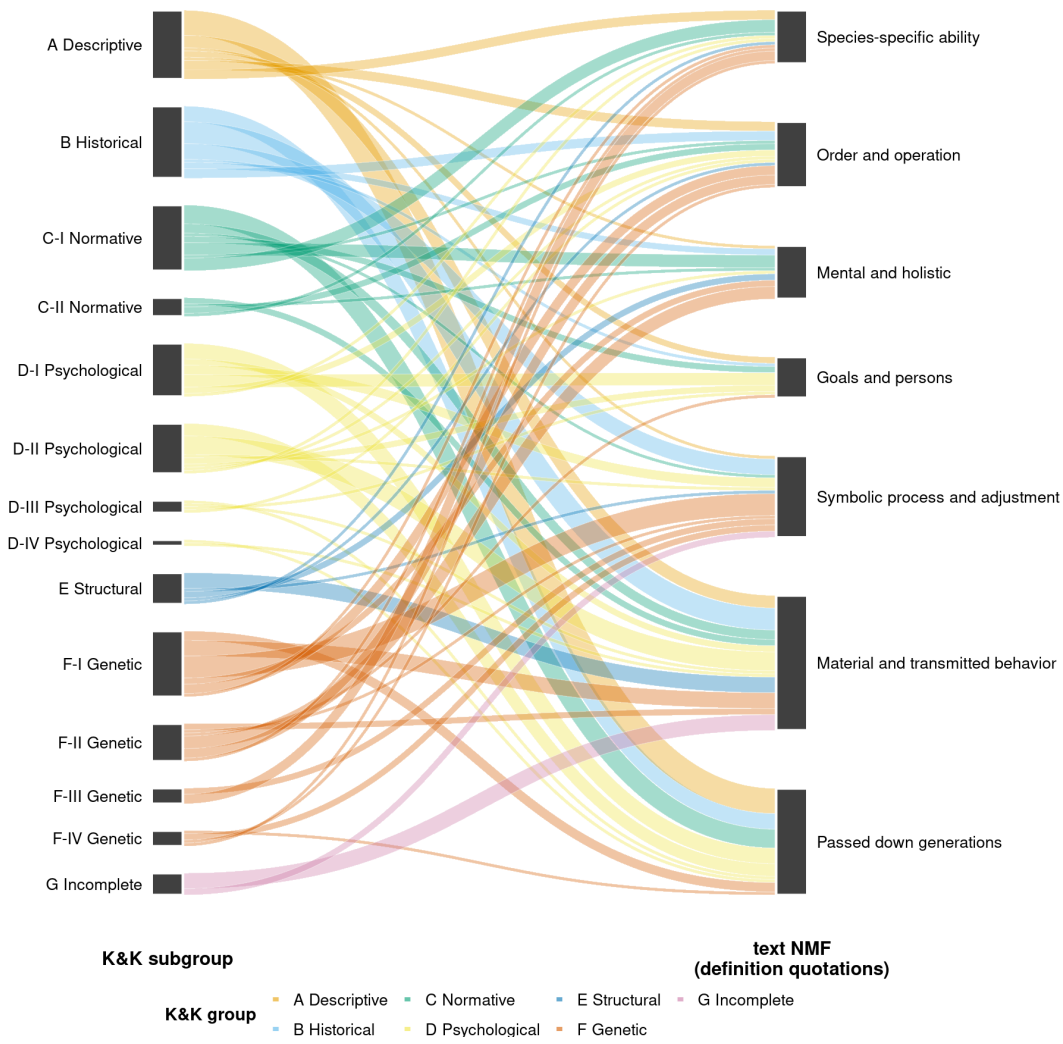


Figure 7: The 171 definition codes carrying a text, by their own K&K subgroup (left) and dominant text NMF factor (right). Ribbons take the color of their origin node's group.

Breiger, R. L. (1974). The duality of persons and groups. *Social Forces*, 53(2):181–190.

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